
PaperMemory: Evidence-Packet Retrieval for Local-First Literature Assistance

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Abstract

PDF-based literature work can be slow because researchers need page-level evidence, citation support, and visible uncertainty before they can trust a draft answer. PaperMemory is a local-first compound AI assistant for postgraduate researchers who work with bounded PDF collections. It builds page-level EvidencePackets, combines visual and selectable-text retrieval, runs a bounded evidence-seeking loop, checks reliability, and keeps human verification in the workflow. Early market evidence from informal interviews with 10 PhD, master's, or postgraduate researchers gives a mean conditional purchase likelihood of 4.2/5 if the claimed functions are delivered, and 70% prefer PaperMemory-provided API/model access through a monthly subscription rather than BYOK-only setup. We therefore evaluate a freemium model with a free BYOK tier and proposed 10 and 20 USD/month tiers that include weekly capped, provider-compliant API relay for managed LLM access. On a controlled five-question cross-document retrieval fixture, all methods are close in mean evidence recall@5, while hybrid page fusion fully supports 3/5 questions versus 1/5 for keyword overlap and 2/5 for each single-retrieval baseline; robustness fixtures cover missing evidence, citation drift, prompt-injection-like PDF text, conflicts, and low-text pages. A cost stress test estimates positive base-case net savings for first-pass evidence gathering under stated assumptions, while leaving conversion, retention, hosting cost, support cost, licensing, and real-corpus retrieval quality as open commercialization risks.

1 Introduction

Postgraduate researchers often spend early writing time turning PDFs into claims they can safely cite. The hard part goes beyond finding a relevant page. A useful research assistant must show why a page was used, which citation supports the sentence, and where the answer should stop because the evidence is thin. When that boundary is hidden, users still have to redo the evidence work by hand.

PaperMemory targets this paid workflow directly. In informal interviews with 10 university PhD, master's, or postgraduate researchers, the mean conditional purchase likelihood was 4.2/5 if PaperMemory delivered the promised functions: local PDF ingestion, evidence-backed answers, visible citations, bounded uncertainty, and a simpler model-access path. Seven of the 10 interviewees preferred PaperMemory-provided API/model access through a monthly subscription over a BYOK-only setup. Appendix B records the short questionnaire protocol and aggregate response framing. This is early willingness-to-pay evidence, not product-market fit, but it is enough to justify testing a paid model in this course-stage report.

Existing research assistants such as Elicit and Consensus are priced around evidence-oriented workflows, although they are market comparators rather than executable baselines for this report (Elicit,

Table 1: Postgraduate interview signal used for the market and revenue story.

Evidence item	Result
Interview sample	10 postgraduate interviewees.
Conditional purchase likelihood	4.2 / 5 mean if the described workflow is delivered.
API/model access preference	70% prefer product-provided API/model access through monthly subscription over BYOK-only setup.
Proposed packaging	Free BYOK plus 10 and 20 USD/month tiers.

Table 2: Major report claims and supporting evidence.

Claim	Evidence in this report
Market signal for a paid postgraduate workflow.	Interview aggregate: n=10 postgraduate researchers, 4.2/5 conditional purchase likelihood, and 70% managed-access preference.
PaperMemory is a compound AI system rather than a chat-only UI.	EvidencePackets, visual and text retrieval paths, BM25, hybrid page fusion, bounded evidence loop, reliability status, and evidence-visible UI.
Retrieval components work on the local fixture suite.	Keyword, visual API, BM25, and hybrid rows; BM25 reaches 88.9% R@1 and 100.0% retrieval-level refusal correctness.
Subscription packaging has bounded cost logic.	10-PDF/30-PDF cost stress test plus provider-compliant API relay sensitivity.
Trust behavior is implemented as visible limits.	Robustness matrix, failure gallery, reliability labels, citation filtering, and prompt-injection-like PDF text handling.

2026; Consensus, 2026). PaperMemory’s narrower position is local-first PDF work: the user brings a scoped collection, the system builds page-level EvidencePackets, and the interface exposes citations, limits, and reliability status before the user turns the draft into research writing.

The contribution is both technical and commercial. Technically, PaperMemory combines page rendering, selectable-text manifests, BM25 retrieval, page-canonical rank fusion, EvidencePacket validation, bounded evidence seeking, reliability checks, and an evidence-visible UI. Commercially, it connects that workflow to a freemium BYOK product and proposed paid tiers that remove API setup friction through compliant managed model access. The evaluation remains bounded: synthetic/local metrics and controlled live tests show implementation behavior, not broad scholarly-corpus superiority.

2 Product and Revenue Logic

The first target users are postgraduate researchers who work with dense PDF collections and need a faster first pass over evidence. A second group is small research, consulting, or R&D teams that need traceable evidence packets before writing memos or reports. PaperMemory is not a systematic-review replacement or final expert adjudicator. It supports evidence gathering, citation packaging, and uncertainty surfacing before human review.

The proposed packaging follows the interview signal. The free tier is BYOK with limited groups, which lowers adoption friction and keeps model cost outside the product. The 10 USD/month tier adds provider-compliant API relay for managed LLM access with a lower weekly managed-model allowance for routine postgraduate work. The 20 USD/month tier offers a larger workspace and a higher weekly managed-model allowance for heavier literature projects. The paid design assumes commercial API access or equivalent provider permission; it does not rely on consumer-account pooling, coding-plan resale, or unlimited inference. The exact quotas should be set only after usage telemetry exists; this report intentionally does not invent them.

A later 50 USD/month organization tier could extend the same evidence workflow to small labs, research groups, or consulting teams by supporting shared literature libraries and shared notes across users. This tier is framed as collaborative workspace value rather than higher individual model usage; its economics would require separate telemetry on seats, storage, collaboration patterns, and support cost.

Table 3: The proposed freemium model connects managed model access with evidence-workflow value.

Tier	User value	Model/API handling
Free BYOK	Limited groups for local PDF evidence work and adoption testing.	User supplies their own key or local model path.
10 USD/month	Lower-friction postgraduate workflow with more groups and routine evidence runs.	Provider-compliant API relay for managed LLM access under a lower weekly cap.
20 USD/month	Larger workspace for heavier literature projects and repeated report drafting.	Provider-compliant API relay under a higher weekly cap and larger workspace.

The value logic is that users pay for convenience and evidence discipline together. The subscription removes API-key setup for the 70% of interviewed users who preferred managed access, while the product keeps the local-first evidence workflow visible: pages, citations, limits, and reliability status remain part of the answer. That distinction matters because a plain hosted chatbot can be cheaper to build, but it does not give the user the same packet-level audit trail.

The cost model supports a plausible subscription experiment under bounded usage. The cost stress test estimates API/compute costs of \$0.19 for a 10-PDF evidence packet, \$0.68 for a 30-PDF project evidence scan, and \$1.62 for systematic-review pre-screening, with positive base-case net savings in all three scenarios after operation, verification, API/compute, and scenario license assumptions. Appendix D records the assumptions behind this stress test. These numbers do not prove margins. They exclude production hosting, support, abuse controls, full licensing decisions, sales cycle cost, usage variance, actual conversion, and retention. The 10 and 20 USD/month tiers are therefore commercially plausible only under fair-use limits, and they remain proposed packaging rather than validated revenue.

3 Compound AI System Architecture

PaperMemory is a compound AI system over a local PDF collection. The implemented stack combines a Qdrant-compatible page-vector store, VisRAG-style page-image retrieval, a BM25 text-manifest index, provider LLM generation through BYOK or managed API relay, server-side EvidencePacket validation, a reliability verifier, and an evidence-visible UI. The ingestion path renders PDFs into page artifacts and extracts selectable text where the PDF permits it. Each page can therefore carry an image, a caption or text snippet, a selectable-text manifest entry, and a quality label such as low-text or OCR-needed. This split matters because visual retrieval can still use page layout, while BM25 can use exact terms when text exists.

Retrieval has three implemented paths. The visual path retrieves page candidates from the page-image evidence surface. The BM25 path indexes the text manifests with $k_1 = 1.5$ and $b = 0.75$. The hybrid path applies page-canonical reciprocal rank fusion with $k = 60$, then collapses visual and text hits onto the same $(paper, page)$ unit. That canonical page key keeps the downstream citation grain stable even when the two retrievers disagree about score scale or rank source.

BM25 is included as a trust-oriented text layer rather than as a novelty claim. It is deterministic, inexpensive, and easy to reproduce, so it gives PaperMemory an auditable evidence surface for exact scientific terms, method names, dataset names, abbreviations, and citation-bearing snippets when selectable text exists (Robertson and Zaragoza, 2009). This matters for commercial trust because a user can inspect the matched page text and understand why the page entered the EvidencePacket, instead of receiving only an opaque embedding score. BM25 therefore supports reproducibility and citation checking even though it is not presented as a new retrieval algorithm.

The visual retrieval layer is motivated by recent state-of-the-art document retrieval work showing that page images can be directly useful for visually rich documents (Yu et al., 2024; Faysse et al., 2024). This design is especially appropriate for academic PDFs because important evidence often appears in figures, tables, equations, page layout, and captions that are degraded or lost when the document is reduced to plain text. VisRAG-style page retrieval lets PaperMemory search the rendered page as the user sees it, while the BM25 layer anchors exact lexical evidence when text

PaperMemory

Local-first evidence retrieval and verified answer generation

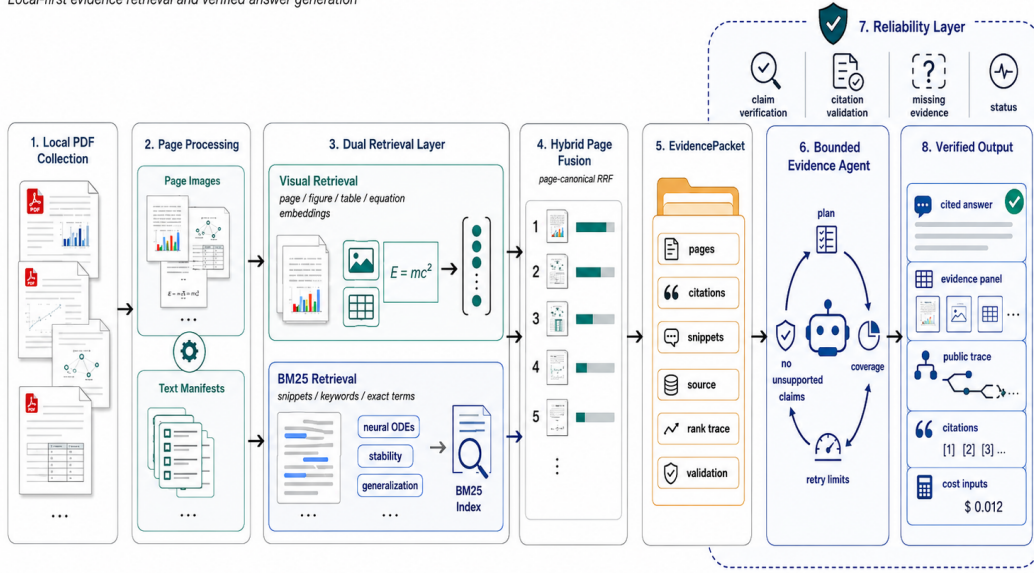


Figure 1: PaperMemory pipeline with the reliability layer. The figure describes the report package over existing local artifacts; it does not add OCR, dense text retrieval, billing, or unrestricted agent actions.

127 extraction is available. The hybrid design is therefore not redundant: it combines visual document
 128 understanding with reproducible text matching before generation is allowed to cite a page.

129 The EvidencePacket schema is the system boundary between retrieval and generation. A packet
 130 records the query, paper scope, accepted EvidenceUnits, accepted page citations, rank traces, source
 131 modality, validation state, and limits. The public packet excludes local filesystem paths and keeps
 132 stable evidence ids so that chat, trace logs, and report material can point back to the same page units.
 133 Generation is allowed to see packet context, but final citations are filtered back to accepted packet
 134 pages.

135 The reliability layer sits after packet construction and before the answer is treated as ready for the
 136 user. It plans evidence requirements from the question, checks whether the selected papers and claim
 137 types are covered, verifies answer claims against accepted evidence, and returns a visible **strong**,
 138 **partial**, or **insufficient** status. The status is a review signal, not a correctness certificate. It
 139 tells the user whether the packet supports the answer, where coverage is thin, and when the system
 140 should withhold or narrow a claim.

141 This architecture also connects to the product cost model. Reliability checks add planner and veri-
 142 fication calls when enabled, while disabled reliability mode avoids that extra work. The cost stress
 143 test therefore belongs in the architecture story as a deployment control: stronger grounding can be
 144 purchased with more model work when the task warrants it, and lighter runs remain available for
 145 lower-risk exploration.

126 4 Agentic Autonomy and Implementation Logic

127 PaperMemory’s autonomy is bounded and server-verified. The agent does not browse freely, execute
 128 arbitrary tools, or attempt a complete literature review. It follows a small state machine over the
 129 selected paper scope. Planner output is treated as an untrusted hint, then the server clamps pass
 130 budgets, query counts, and final evidence size before retrieval or answer generation can proceed.

131 The implemented loop starts with query rewrite. The system prepares a conversation-aware retrieval
 132 query, then runs first retrieval through the selected retrieval mode, normally hybrid for scoped chat.
 133 Evidence analysis inspects the accepted packet units and records missing requirements. If coverage

Table 4: Bounded agent trace states used by the evidence-seeking loop.

State	Agent action	Tool or component	Stop or transition
Rewrite	Prepare a scoped retrieval query from the user turn.	Server planner hint.	Move to first retrieval.
Retrieve	Retrieve page evidence for the selected papers.	Hybrid, visual, or BM25 retrieval.	Build accepted EvidencePacket units.
Analyze	Compare packet evidence with required claims.	Requirement planner and coverage check.	Stop as <code>sufficient</code> or request targeted retry.
Retry	Search for uncovered requirements.	Coverage-gated retrieval retry.	Add only new accepted page units.
Check	Check whether retry changed the packet.	Evidence delta and dedupe check.	Stop as <code>no_new_evidence</code> when delta is zero.
Answer	Generate with final packet context and cite accepted pages.	Chat service and claim verifier.	Return answer plus reliability status.

Table 5: Realistic five-question cross-document retrieval comparison. Higher is better.

Method	Mean evidence recall@5	Fully supported questions
Keyword overlap baseline	0.80	1/5
VisRAG page-image retrieval	0.81	2/5
BM25 text-manifest retrieval	0.82	2/5
Hybrid page fusion	0.85	3/5

is incomplete and the budget remains, the orchestrator issues a targeted second retrieval rather than answering immediately. The retry is coverage-gated: it must seek missing evidence, deduplicate accepted units, and report whether the new pass changed the packet.

Stop reasons are explicit. A `sufficient` trace can answer after the first pass when the packet covers the question. A `no_new_evidence` trace stops after a retry produces no new accepted units. The answer step then uses the final packet, strips citations that do not match accepted pages, and attaches the reliability status from the requirement, coverage, and claim-verification checks. A `partial` answer can still be useful when it says which part is supported. An `insufficient` answer asks the user to revise scope or add evidence instead of filling the gap with unsupported prose.

The live controlled run exercises the same boundary. The successful image-context chat returned HTTP 200, six evidence rows, six citations, three included evidence images, and a final `sufficient` stop reason. The missing-evidence chat used a bounded second retrieval and ended with `no_new_evidence`, while the answer stated that the evidence packet did not support the requested mitochondrial ribosome profiling claim. The prompt-injection-style page stayed evidence rather than instruction: the answer treated the page text as untrusted and did not reveal keys, execute tools, or alter citations. Appendix C lists the essential trace artifacts behind this run.

5 Baselines, Evaluation, and Technical Results

The evaluation is a proof-of-capability package built from synthetic fixtures, deterministic regression tests, and a controlled local browser/API run. It is not a broad benchmark over scholarly PDFs. The synthetic retrieval suite has four papers, 12 pages, 12 golden questions, and three negative cases. The controlled local run used uploaded fixture PDFs and records UI flow, retrieval quality checks, bounded agent traces, and trust behavior. These artifacts support implementation claims, not claims about general corpus performance.

Table 5 reports a realistic five-question cross-document retrieval fixture derived from the skill-security demo corpus. Unlike the earlier page-lookup sanity check, these questions ask for cross-paper comparison, numeric grounding, taxonomy use, defense reasoning, and report-facing claim scoping. Mean recall values are close across methods, so the more interpretable signal is whether a method retrieves every required support page for a question. Keyword overlap fully supports 1/5 questions, single-modality VisRAG page-image retrieval and BM25 text-manifest retrieval support

Table 6: Deterministic robustness cases covered by local regression fixtures.

Case	Expected behavior
Empty scope	Conversation-mode limit and paper-style citation stripping.
Missing evidence	Partial packet with no accepted citations.
No-new-evidence loop	Stop reason records zero evidence delta.
Prompt-injection-like PDF text	PDF text remains untrusted evidence; unsafe planner hints are redacted.
Citation drift	Unsupported page labels are stripped.
Conflicting numeric evidence	Verification-needed limit is surfaced.
Low-text marker	Visual-first limit warns against invented missing text.

2/5 each, and PaperMemory’s hybrid page fusion supports 3/5 by combining page-level visual cues with exact text evidence before citations are passed to the answer layer. Appendix C records the per-question fixture output.

The realistic retrieval fixture remains controlled: it uses page aliases and anchor text derived from the five-question demo corpus rather than a broad scholarly PDF benchmark. Its purpose is to test the retrieval pattern used by PaperMemory’s product workflow: a user asks a natural question, the system retrieves relevant pages from selected papers, and the answer layer receives page-grounded evidence. The separate live five-question run remains an answer-quality smoke test with partial manual quality; Table 5 isolates retrieval coverage so the method comparison is clean.

The bounded evidence agent is evaluated through traces and regression behavior rather than Recall@k. The trace examples show accepted evidence ids, pass indexes, evidence deltas, stop reasons, and final packet limits. The controlled live run adds a more realistic smoke test: search and chat reached the local API, evidence cards appeared in the UI, citation chips were rendered, missing evidence stopped safely, and prompt-injection-style PDF text remained inside the evidence boundary. The main technical result is therefore narrower than a benchmark claim: the current implementation connects retrieval, EvidencePackets, bounded retries, reliability status, and UI evidence surfacing in a reproducible local workflow.

6 Trust, Safety, and Robustness

Trust in PaperMemory depends on visible failure behavior. A research assistant can be useful only if the user can see when the evidence packet is strong, partial, or insufficient. PaperMemory therefore treats every answer as a product of selected papers, accepted EvidenceUnits, packet limits, and a reliability status. The status is a review signal for the user. It is not a correctness certificate.

The robustness package covers common ways a PDF-grounded assistant can fail. With an empty scope, PaperMemory enters conversation mode and strips paper-style citations because no selected paper evidence exists. With missing scoped evidence, the packet is partial, carries no accepted citations, and removes generated page labels that are not supported by the accepted packet. With repeated retrieval, a second pass that finds the same page stops with `no_new_evidence` rather than treating a duplicate as new support.

Prompt-injection-like PDF text is handled as untrusted evidence. The page text may appear inside the accepted evidence section, but planner prompts label it as evidence rather than instruction, extra action fields are rejected, and unsafe planner hints are removed from public traces. Citation drift is handled at serialization time: citations outside accepted packet pages, including unknown-paper labels, are stripped before the answer reaches the user. Conflicting numeric evidence adds a verification-needed limit instead of forcing a single value. Low-text, empty, or OCR-needed pages add a visual-first limit so the assistant does not invent text that was not extracted.

The reliability layer turns these cases into user-facing states. A `strong` answer has the planned evidence requirements covered by accepted pages. A `partial` answer can be useful, but the limits explain which evidence is missing or contested. An `insufficient` answer tells the user to change scope or add sources before treating the answer as paper-grounded. This is a practical review workflow for controlled local fixtures and live smoke tests. It is not a broad security result, an

Table 7: Cost stress-test estimates under stated assumptions. API/compute cost is per scenario, and savings are net USD after operation, human verification, API/compute, and license assumptions.

Scenario	PDFs	API/compute	Low savings	Base savings	High savings
10-PDF evidence packet	10	0.19	52.35	183.70	334.76
30-PDF project evidence scan	30	0.68	79.32	561.61	1175.45
Systematic-review pre-screening	200	1.62	-232.20	3423.86	7863.37

Table 8: Proposed pricing tiers for a subscription experiment.

Tier	Access model	Human role
Free BYOK	User provides an API key or local model path for limited local PDF groups; no included model spend.	User verifies pages and manages their own model setup.
10 USD/month	Provider-compliant PaperMemory API relay for users who prefer subscription access over BYOK-only setup.	User verifies cited pages before using claims in writing.
20 USD/month	Provider-compliant API relay with a higher weekly managed-model allowance.	User remains responsible for expert judgment and final citation checks.

OCR result, or proof that every unsupported claim will be detected. Appendix D summarizes the robustness fixtures and cost assumptions.

7 Commercial Stress Test

The commercial question is whether evidence-packet work can save enough expert time to justify a paid workflow while keeping human verification in place. The stress test models that question with:

$$\text{net savings} = (h_m - h_p - h_v)r - c_a - c_l,$$

where h_m is manual work, h_p is PaperMemory operation time, h_v is human verification time, r is a fully loaded hourly rate, c_a is API or compute cost, and c_l is a license allocation. The formula makes verification a cost, not an optional afterthought.

The per-scenario API or compute cost is small in the current stress test: 0.19 USD for a 10-PDF evidence packet, 0.68 USD for a 30-PDF project evidence scan, and 1.62 USD for systematic-review pre-screening. Base-case net savings are 183.70 USD, 561.61 USD, and 3423.86 USD after operation, verification, API/compute, and license assumptions. The low-savings case for systematic-review pre-screening is -232.20 USD, which matters. It shows that the value case can fail when verification time and deployment assumptions dominate.

The interview signal supports a subscription experiment rather than a BYOK-only product. Seven of 10 interviewed postgraduate researchers preferred PaperMemory-provided API/model access through a monthly subscription. Table 9 uses the 10-PDF and 30-PDF cost-stress-test estimates as illustrative managed-access cases, not production quotas or validated margins. The proposed packaging keeps the free tier as BYOK, then tests 10 USD/month and 20 USD/month managed-access tiers through compliant API relay. Those tiers are plausible only with weekly capped managed LLM usage, BYOK fallback, rate limits, or a fair-use policy. Exact quotas should come from usage telemetry, not this report.

The paid tiers bundle compliant managed LLM access support rather than unlimited inference. The weekly cap is the key margin control: once a user exceeds the managed allowance, the product can pause, ask the user to wait, or fall back to BYOK. This keeps the commercial claim tied to a bounded service design rather than an unlimited API subsidy or subscription-account arbitrage. The stress test does not prove paid conversion, retention, production margins, hosting cost, support cost, abuse cost, or legal cost. It also does not make PaperMemory a systematic-review replacement. The stronger commercial claim is narrower: under the current assumptions, API cost is not the main blocker for first-pass evidence gathering. The product risk sits in user adoption, verification effort, usage variance, and deployment licensing.

Table 9: Illustrative subscription stress test for provider-compliant managed-LLM API relay.

Tier	Modeled month	API cost and headroom
Free BYOK	User-supplied model/key.	0 USD PaperMemory model spend; 0 USD subscription revenue.
10 USD/month	Four 10-PDF evidence-packet runs.	$4 \times 0.19 = 0.76$ USD; 9.24 USD before non-model costs.
20 USD/month	Four 30-PDF evidence-scan runs.	$4 \times 0.68 = 2.72$ USD; 17.28 USD before non-model costs.
Heavy-use guard	Eight 30-PDF evidence-scan runs.	$8 \times 0.68 = 5.44$ USD; 14.56 USD before non-model costs.

8 Limitations, Ethics, and Reproducibility

The evaluation package is local and controlled. Retrieval metrics come from synthetic fixtures, not a broad scholarly PDF benchmark. The live test uses a small controlled corpus and includes one duplicate-upload pollution caveat. Text extraction is selectable-text only; low-text and scanned pages are flagged, but OCR is future work. The text path uses BM25 and page-canonical fusion, not dense semantic text retrieval. The reliability layer checks packet support for answer claims, but it does not establish exhaustive literature coverage.

The market evidence is also early. The interview sample has 10 postgraduate researchers, a mean conditional purchase likelihood of 4.2/5, and a 70% preference for managed API/model access. The appendix records the questionnaire protocol, but the report uses only aggregate notes. Those numbers justify a product experiment, not a claim about paid adoption. The planned 10 USD/month and 20 USD/month tiers still need pricing tests, usage telemetry, support estimates, and churn data.

Ethically, PaperMemory should remain a first-pass assistant. It can reduce citation packaging work, but users still need to inspect cited pages before academic, medical, legal, or policy use. The local-first and BYOK options reduce some centralization risk, yet external model APIs may still receive prompts and evidence snippets if the user configures them. Public traces therefore avoid local paths and unsafe planner hints, and generated citations are constrained to accepted packet pages.

Reproducibility materials live under the PaperMemory repository root. Appendix A records the branch, baseline environment, LaTeX command, and verification snapshots. Appendix B records the interview sample, questionnaire protocol, aggregate results, and limits. Appendix D records the robustness gallery and cost assumptions. A separate three-minute road-show video accompanies the submission and shows the investor-facing flow: PDF upload, evidence retrieval, a reliability warning for unsupported claims, and the subscription value proposition; the report focuses on technical evidence, market proof, cost stress testing, and reproducibility.

9 Conclusion

PaperMemory addresses a concrete research workflow: turning a local PDF collection into cited first-pass evidence that a human can inspect. The current system is more than a chat box because it combines visual and text retrieval, page-canonical fusion, EvidencePackets, bounded evidence seeking, reliability status, and UI evidence surfacing. The strongest evidence is still local: synthetic retrieval fixtures, deterministic robustness tests, a controlled live run, early postgraduate interviews, and a cost stress test.

The next steps are practical rather than rhetorical. PaperMemory needs real local-corpus evaluation, OCR and dense text retrieval experiments, usage telemetry for the proposed subscription tiers, and clearer licensing decisions before deployment. Until then, the right claim is bounded: PaperMemory is a plausible local-first assistant for first-pass evidence gathering and citation packaging, with visible limits and human verification built into the workflow.

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301 A Reproducibility Commands and Environment

302 The report package is built from the PaperMemory repository root. The recorded environ-
303 ment for the final package is branch `codex/papermemory-evidence-agent`, baseline commit
304 `6884d738df9d685bad4f5a673a7d13a7bc2fb691`, Python 3.12.7, Node.js v22.18.0, and npm
305 10.9.3.

306 The final report package was compiled with:

```
307 cd reports/final
308 tectonic -X compile --outdir . --outfmt pdf --untrusted main.tex
```

309 The report uses only a small set of local evidence artifacts. They are kept out of References be-
310 cause they are project evidence rather than external literature. Unless noted otherwise, filenames in
311 Table 10 live under `reports/final/results/`.

312 Additional audit artifacts remain in the repository, but the table above lists the evidence objects
313 needed to read and reproduce the report’s major claims.

314 The package uses the official `neurips_2025.sty` downloaded from the NeurIPS 2025 style link
315 on the Call for Papers page (NeurIPS, 2025a). The report uses anonymous submission style, not
316 camera-ready or preprint mode.

Table 10: Essential local evidence artifacts used by the report.

Evidence need	Essential artifact	Main claim supported
Market validation	Interview appendix	Aggregate interview protocol, 4.2/5 conditional purchase likelihood, and 70% managed-access preference.
Retrieval comparison	Cross-document metrics	Five-question cross-document retrieval table and per-question evidence pages.
Agent behavior	Live trace logs	Bounded retrieval retry, supported/insufficient answer states, and citation filtering behavior.
Robustness and cost	Robustness/cost files	Safe-failure fixtures and commercial stress-test assumptions.
Build and style	Build notes/style hash	Compile command, package layout, and NeurIPS style file used for the PDF.

B Interview Market Validation

The market evidence comes from informal interviews with 10 postgraduate interviewees. The sample is small, so the results are treated as early intent evidence rather than paid adoption data.

Questionnaire protocol. Each interview used the same six prompts:

1. What is your current research stage, and how often do you read PDFs?
2. What is your current literature reading and citation-checking workflow?
3. On a 1–5 scale, how likely would you be to pay if PaperMemory delivered local PDF ingestion, evidence-backed answers, visible page citations, bounded uncertainty, and smoother model access?
4. Would you prefer BYOK/local-model setup or PaperMemory-provided LLM access through a subscription?
5. Which packaging would you prefer: free BYOK with limited groups, 10 USD/month with a lower weekly managed-model limit, or 20 USD/month with a higher weekly managed-model limit?
6. What risk would stop you from relying on the tool for research writing?

Aggregate response. Across the 10 postgraduate interviewees, the mean conditional purchase likelihood was 4.2/5 if PaperMemory delivered the described workflow, and 70% preferred PaperMemory-provided API/model access through a subscription over BYOK-only access. The report frames that paid option as provider-compliant API relay with caps, not consumer-account resale. The report does not claim participant-level raw data, paid conversion, retention, product-market fit, or production margins from this interview sample.

Table 11: Interview aggregate used in the market and revenue sections.

Measure	Result	Report use
Sample	10 postgraduate researchers	Early postgraduate researcher signal, not a representative survey.
Conditional purchase likelihood	4.2 / 5 mean	Supports testing a paid evidence-workflow product if the promised functions are delivered.
Managed API/model preference	70%	Supports adding monthly subscription tiers rather than offering BYOK only.
Proposed tiers discussed in report	Free BYOK, 10 USD/month, 20 USD/month	Proposed packaging for stress testing, not launched billing.

How the report uses the result. The main report uses the interviews to justify a commercial experiment: a free BYOK tier for adoption, plus paid provider-compliant managed-access tiers for users who do not want to configure model keys. The result also explains why the cost model matters. If managed access is part of the product, model usage must have weekly caps, BYOK fallback above cap, rate limits, or fair-use controls.

Limitations. The interviews do not measure paid conversion, retention, referral, production revenue, support load, or willingness to keep paying after repeated use. The report therefore treats the numbers as market direction, not proof of demand.

C Essential Evaluation Details

The main retrieval table is generated by:

```
python eval/retrieval/run_retrieval_eval.py --mode realistic-crossdoc \
--output-prefix realistic-crossdoc
```

The fixture uses five user-facing questions over the skill-security demo corpus. The generated Markdown and JSON artifacts store the per-question expected pages, retrieved evidence pages, recall@5, and full-support flags.

Table 12: Essential retrieval metrics restated from the generated artifact.

Method	Mean evidence recall@5	Fully supported questions
Keyword overlap baseline	0.80	1/5
VisRAG page-image retrieval	0.81	2/5
BM25 text-manifest retrieval	0.82	2/5
Hybrid page fusion	0.85	3/5

The report-facing table is stored as `realistic_crossdoc_metrics.md` under `reports/final/results/`. The machine-readable per-question rows are stored as `realistic-crossdoc.json` under `eval/retrieval/results/`. The same command also writes CSV and Markdown summaries beside the JSON file.

The live local smoke evidence is intentionally summarized rather than reproduced in full. The essential records are `live_local_test_log.md` and `agent_trace_examples.md` under `reports/final/results/`. These files support the bounded-agent claims in the main text: first-pass sufficient answers, missing-evidence retry, `no_new_evidence` termination, citation filtering, and prompt-injection-like PDF text staying inside the evidence boundary.

D Robustness Failure Gallery and Cost Assumptions

The failure gallery records seven safe-failure snapshots: empty scope, missing scoped evidence, no-new-evidence loop, prompt-injection-like PDF text, citation drift, conflicting evidence, and low-text or scanned markers. The robustness matrix records 11 focused test passes for these fixtures. The expected behavior is containment and visible limits, not complete prevention of all possible prompt or citation failures. The supporting local artifacts are `robustness_matrix.md` and `failure_gallery.md` under `reports/final/results/`.

The cost model uses the stress-test formula:

$$\text{net savings} = (h_m - h_p - h_v)r - c_a - c_l.$$

Low-savings, base-savings, and high-savings cases vary manual and verification hours. Labor conversion uses annual wage divided by 2080 hours and multiplied by 1.35. Source values for OpenAI pricing, BLS wages, Elicit, and Consensus are a 2026-06-24 planning snapshot, so they should be refreshed before any customer-facing or investor-facing use.

PyMuPDF/MuPDF licensing remains unresolved for commercial deployment. The cost table therefore includes planning license allocations, but the paper does not convert those estimates into

Table 13: Cost-model assumptions from `cost_benefit.csv`.

Scenario	Runs	Input tok.	Cached tok.	Output tok.	Labor profile
10-PDF evidence packet	3	175000	25000	17500	BLS management analysts, 65.68 USD/hr loaded.
30-PDF project evidence scan	6	690000	120000	54000	BLS data scientists, 73.08 USD/hr loaded.
Systematic-review pre-screening	10	3500000	750000	250000	BLS medical scientists, 65.29 USD/hr loaded.

production margins. The cost spreadsheet and narrative are stored as `cost_benefit.csv` and `cost_benefit.md` under `reports/final/results/`.

The proposed 10 USD/month and 20 USD/month tiers are tier-economics assumptions rather than validated pricing telemetry. Weekly caps, BYOK fallback above cap, rate limits, and fair-use controls are product controls for provider-compliant managed LLM API relay; the report does not assume consumer-account pooling, unlimited inference, or production margins.

NeurIPS Paper Checklist

This checklist follows the NeurIPS checklist guidance (NeurIPS, 2025b). Answers are intentionally conservative.

- Claims.** [Yes] The abstract and introduction state that performance metrics are synthetic/local smoke evidence, and that interview evidence is early conditional intent rather than paid adoption.
- Limitations.** [Yes] Section 8 and Appendix D cover synthetic fixtures, no OCR, no dense text retrieval, no broad user study, and licensing risk.
- Theory.** [NA] The paper does not introduce theorems or formal proofs.
- Reproducibility.** [Yes] Appendix A records commit, environment, commands, and local artifact paths.
- Datasets.** [Yes] The evaluation data are synthetic local fixtures described in the metrics artifacts. Appendix B describes the small interview sample used for market proof.
- Compute.** [Yes] The local Python, Node, npm, and LaTeX verification commands are recorded in the build notes and appendix.
- Ethics.** [Yes] Section 8 discusses human verification, privacy, prompt-boundary limits, and licensing risk.
- Broader impacts.** [Yes] The system can reduce first-pass citation labor and API setup friction, but misuse risk remains if users treat draft answers as verified scholarship.
- Safeguards.** [Yes] Deterministic guardrails are described for empty scope, missing evidence, prompt-injection-like PDF text, citation drift, conflicting evidence, and low-text markers.
- Licenses and assets.** [Yes] Local artifacts are listed, and PyMuPDF/MuPDF AGPL/commercial-license exposure is documented as a deployment risk.